Topic 5: Digital Modulation

CSE 4405: Data and Telecommunications

Md. Tanvir Hossain Saikat, Jr. Lecturer

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Islamic University of Technology (IUT)

Learning Objectives

1. Explain why digital communication over band-pass channels requires carrier modulation
2. Relate bit rate, symbol rate, and modulation order (M-ary signaling)
3. Build intuition for ASK, FSK, PSK, DPSK, QAM, and continuous-phase FSK families
4. Interpret constellation diagrams and decision regions without memorizing isolated formulas
5. Explain BER/SER behavior using symbol spacing and E_b/N_0 intuition
6. Connect modulation choices to real systems and to the rest of CSE 4405

Reading focus: Forouzan 5e, Ch. 5 (digital-to-analog conversion and digital modulation family, typically pp. 151–166 in common printings); Goldsmith Ch. 5 (pp. 126–171); Rappaport 2e Ch. 5–6 (digital passband and receiver perspective).

Why We Need Carrier Modulation

Central Question

Why can't we just send digital bits directly over wireless and wired band-pass channels?

- Base-band digital signals (pulses near DC) have energy centered at low frequencies.
- Most practical channels are **band-pass**: they pass signals only in a range $[f_L, f_H]$.
- Radio, wireless, and telephone line channels are band-pass (frequency-selective).
- Sending a low-frequency digital signal through a band-pass channel causes severe attenuation.

Fourier View: Why Direct Digital Fails

A baseband digital waveform is a sum of pulses. For a symbol pulse $p(t)$, Fourier analysis gives a spectrum $P(f)$ concentrated around $f = 0$.

$$\text{rectangular pulse:} \qquad P(f) = T_s \, \mathrm{sinc}(fT_s)$$

Directly through a band-pass channel

```
D(f) = data signal spectrum
(fourier transformation of the
transmitted data waveform)
H(f) = tells what the channel
does to each freq (large / near
1 -> channel passes the freq)
(small / near 0 -> channel
blocks the freq)
H(f) ≈ 0 for low freqs
```

$$Y(f) = D(f)H(f)$$

- $D(f)$: data spectrum near DC
- $H(f) \approx 0$ near DC
- Main pulse energy is removed or badly distorted

After carrier modulation

$$D(f) \to \frac{1}{2}\left[D(f - f_c) + D(f + f_c)\right]$$

- The spectrum is shifted into the allowed band.
- The receiver shifts it back down and filters it.

Reason: a band-pass channel is not just "wide enough"; its allowed frequencies must overlap the signal spectrum. Carrier modulation creates that overlap.

Base-Band vs. Band-Pass Channels

Base-Band Channel

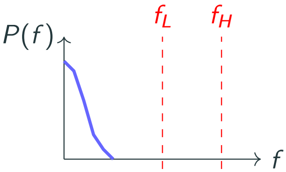
- Passes DC and low frequencies.
- Example: short copper links designed for baseband line coding.
- Digital pulses can travel directly.
- Bandwidth: 0 to $f_{\max}$.

Band-Pass Channel

- Passes signals only in a frequency range $[f_L, f_H]$.
- Example: Wireless, radio, telephone lines.
- Blocks DC and very low frequencies.
- Bandwidth: $(f_H - f_L)$.

Channel Mismatch Problem

Problem: Digital pulses have most energy near DC, but band-pass channels pass only a nonzero frequency band.



Solution: Shift the spectrum to center it in the band-pass region using a carrier.

Carrier Modulation: The Mechanism

A carrier signal is a high-frequency sinusoid:

$$c(t) = A_c \cos(2\pi f_c t + \phi)$$

- A_c: carrier amplitude
- f_c: carrier frequency (chosen at the band-pass center)
- ϕ: carrier phase

Modulation multiplies the base-band data $d(t)$ by the carrier:

$$s(t) = d(t) \cdot c(t) = d(t) \cdot A_c \cos(2\pi f_c t + \phi)$$

This **shifts** the base-band spectrum from DC to f_c, fitting it into the band-pass region.

Frequency Domain: Why Modulation Works

Modulation Property: If $d(t)$ has spectrum $D(f)$, then $d(t) \cdot \cos(2\pi f_c t)$ has spectrum shifted to f_c.

1. Base-band signal $d(t)$: spectrum near DC.
2. After multiplying by carrier: spectrum centered at $\pm f_c$.
3. Band-pass channel passes the shifted lobes inside $[f_L, f_H]$ (one-sided view: around $+f_c$).
4. Receiver demodulates: multiply by carrier again, recover $d(t)$.

Summary: Why Carrier Modulation Is Essential

- **Band-pass channels are ubiquitous:** Wireless, radio, satellite, telephone lines.
- **Direct base-band transmission fails:** Main low-frequency pulse energy is blocked by the band-pass channel.
- **Carrier modulation solves the problem:** Maps base-band digital data onto a sinusoid at the channel center frequency, placing the signal energy in the pass band.
- **Receiver demodulation:** Multiply by the carrier to shift the spectrum back to base-band and recover the bits.

Key Takeaway

Carrier modulation is not optional for band-pass channels; it is the fundamental technique that places digital information inside the usable passband.

Receiver Demodulation: Mathematical View

Step 1: Transmitted signal (after band-pass channel):

$$r(t) = d(t) \cdot A_c \cos(2\pi f_c t)$$

Step 2: Receiver multiplies by the carrier:

$$r(t) \cdot \cos(2\pi f_c t) = d(t) \cdot A_c \cos^2(2\pi f_c t)$$

Step 3: Use identity $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$:

$$\frac{A_c}{2} d(t) + \frac{A_c}{2} d(t) \cos(4\pi f_c t)$$

Step 4: Low-pass filter removes high-frequency term at $2f_c$:

$$\text{After LPF:} \quad \frac{A_c}{2} d(t)$$

Step 5: Normalize by amplifying by $\frac{2}{A_c}$:

$$\frac{2}{A_c} \cdot \frac{A_c}{2} d(t) = d(t)$$

5.1 Digital-to-Analog Conversion

Definition

Digital-to-analog conversion maps digital data onto a sinusoidal carrier by changing one or more carrier characteristics.

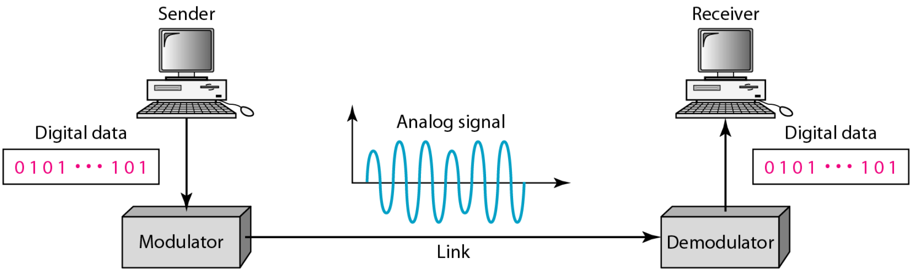
Aspects covered in this section

- Amplitude Shift Keying (ASK)
- Frequency Shift Keying (FSK)
- Phase Shift Keying (PSK)
- Quadrature Amplitude Modulation (QAM)

5.1 Digital-to-Analog Conversion: Block View

Figure: Digital-to-analog conversion

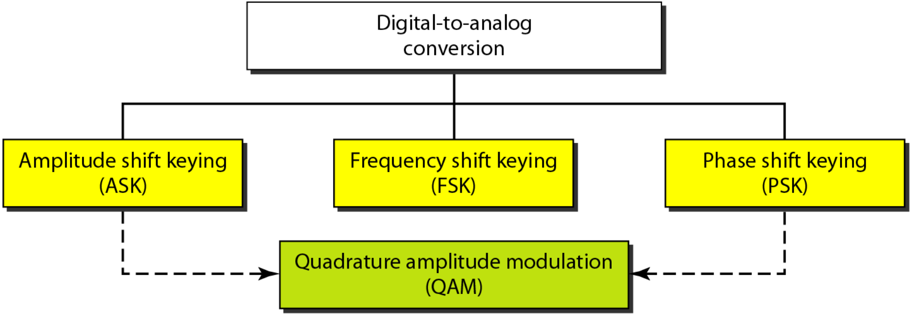
- Bits are converted to carrier symbols at the transmitter.
- A band-pass channel carries the modulated waveform.
- The receiver demodulates and recovers bits.



Digital-to-Analog Conversion (cont'd)

Figure: Type of digital-to-analog encoding

- ASK, FSK, and PSK are basic single-parameter methods.
- QAM combines amplitude and phase.
- Higher-order schemes carry more bits per symbol.



Data Rate vs. Signal Rate

- **Bit rate** N: bits per second (bps)
- **Signal rate** S: symbols per second (baud)
- If one symbol carries r bits, then $S = \frac{N}{r}$
- For L distinct symbol choices (levels, tones, or phases): $r = \log_2 L$
- In digital-to-analog modulation, $S \leq N$

$$N = S \times r, \qquad S = N \times \frac{1}{r}$$

Carrier signal/frequency: the sinusoidal base on which information is mapped.

Example 5.1

Problem: An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find baud rate and bit rate.

- Given $r = 4$, $S = 1000$ baud
- Baud rate $= S = 1000$ baud
- Bit rate $N = S \times r = 1000 \times 4 = 4000$ bps

Result

$S = 1000$ baud, $N = 4$ kbps.

Example 5.2

Problem: An analog signal has bit rate 8000 bps and baud rate 1000 baud. How many data elements are carried by each signal element? How many signal levels are needed?

- Given $N = 8000$, $S = 1000$
- $r = \frac{N}{S} = \frac{8000}{1000} = 8$ bits per symbol
- $L = 2^r = 2^8 = 256$ signal levels

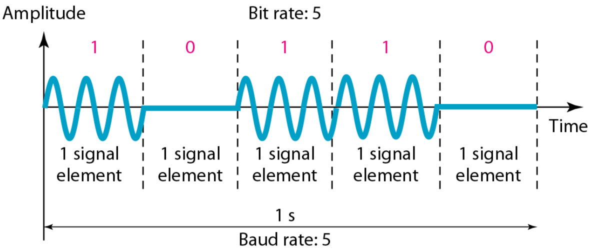
Result

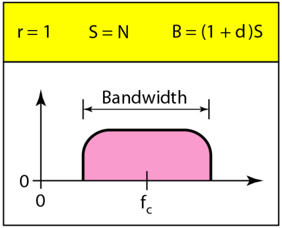
Each signal element carries 8 bits; required levels $L = 256$.

ASK (Amplitude Shift Keying)

used when we want to send digital bits.
contrary to AM / PM / FM when we where sending an analog signal by shifting it to a higher frequency using a carrier wave

- Carrier amplitude changes with data.
- Carrier frequency and phase stay fixed.
- Simple implementation, but vulnerable to amplitude noise.

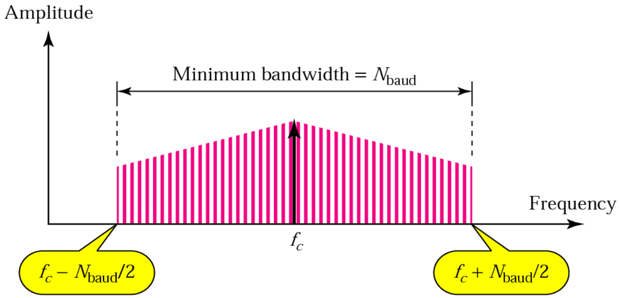




ASK Bandwidth

ASK occupies a passband around f_c.

- Data changes the carrier amplitude at symbol rate $S = N_{\text{baud}}$.
- The spectrum is centered at f_c, not at DC.
- Practical filters add transition bandwidth around the ideal spectrum.



$$B_{\text{ASK}} = (1 + d)S, \qquad 0 \le d \le 1$$

d is the roll-off factor. $d = 0$ gives the ideal minimum $B = S$.

Why Baud Maps to Bandwidth

Time-domain reason

- Symbol period is $T_s = \frac{1}{S}$.
- Larger S means shorter symbols and faster waveform changes.
- Fourier analysis: faster changes require higher-frequency components.
- Therefore the occupied bandwidth B grows with symbol rate S.

Case	Meaning
$d = 0$	$B = S$: ideal minimum, abrupt cutoff, not physically realizable exactly.
$d = 0.5$	$B = 1.5S$: moderate transition region.
$d = 1$	$B = 2S$: widest transition region, easiest filtering.

Takeaway

Baud is not hertz, but baud sets how quickly the waveform changes. The channel needs enough hertz to carry those changes; d adds transition bandwidth.

Pulse Shaping

The physical signal we transmit has Fourier components automatically. Fourier analysis is just our way of describing what frequencies are present in the waveform. In ASK, it is a sine wave that is being turned on and off: s(t)=a(t)cos(2πfc t) where a(t) is like: 1, 0, 1, 1, 0 So the carrier is multiplied by a digital envelope. That sudden turn-on or turn-off is a sharp edge. A waveform with sharp edges naturally requires high-frequency components. You do not intentionally create them; they are just part of the signal's spectrum. A pure sine wave at f_c has only one frequency: f_c. Pulse shaping smooths the switching, so the transmitted ASK signal does not spread too far outside its assigned bandwidth.

Problem: rectangular pulses

- Symbols are sent as abrupt amplitude changes.
- Sharp edges create strong high-frequency components in Fourier analysis.
- The spectrum becomes very wide and can spill outside the assigned channel.

Solution: shape the pulse

- Replace the rectangular pulse with a smoother pulse $p(t)$.
- Smooth pulses have spectra that decay faster.
- A raised-cosine shape controls bandwidth while preserving symbol decisions.

Key Idea

Pulse shaping chooses $p(t)$ so its spectrum $P(f)$ fits the allowed bandwidth and the receiver can still sample symbols cleanly.

Roll-Off Factor d

In this context, when d=0, the filter has an ideal abrupt cutoff in frequency:
B=S
That sounds good because it uses minimum bandwidth, but an infinitely sharp cutoff in frequency causes a very spread-out response in time. So when one symbol is sent, its pulse does not die away quickly. It has long oscillating tails. So with long ringing, one symbol's pulse can spill into neighboring symbol times, which can cause intersymbol interference, or ISI. That means the receiver may have trouble deciding whether it received a 0 or 1 because the previous symbol is still "echoing" into the current symbol.

d tells us how wide the filter transition band is.

$$B_{\mathrm{ASK}} = (1 + d)S$$

d	Spectrum near the band edge	Bandwidth	Practical meaning
0	Abrupt cutoff at $f_c \pm S/2$	S	Ideal minimum; cannot be implemented exactly and causes long ringing.
0.5	Moderate taper between passband and stopband	$1.5S$	Balanced example: more bandwidth, easier filtering.
1	Wide, gradual taper	$2S$	Smoothest transition; uses twice the ideal minimum bandwidth.

Interpretation

Roll-off is not extra data. It is extra spectrum used to make the pulse-shaping filter physically practical.

Bandwidth vs. Filtering Tradeoff

ISI occurs because a real transmitted “symbol” is not perfectly confined to its own time slot. Even if we intend one symbol to last for one symbol period T_s, the actual pulse after filtering/channel transmission usually spreads out in time. So part of symbol k’s waveform is still present when the receiver is trying to sample symbol k+1, k+2, etc. That leftover contribution is intersymbol interference.
A raised-cosine pulse is a special pulse shape used in digital communication to control bandwidth while avoiding ISI at the sampling instants. It allowa symbols to overlap in time, but make sure each symbol contributes zero at the other symbols’ sampling times. That means at the exact decision time for one symbol, the other symbols do not interfere.

Raised-cosine pulses can satisfy zero ISI at sampling instants. The practical tradeoff is bandwidth versus filter sharpness and time-domain ringing.

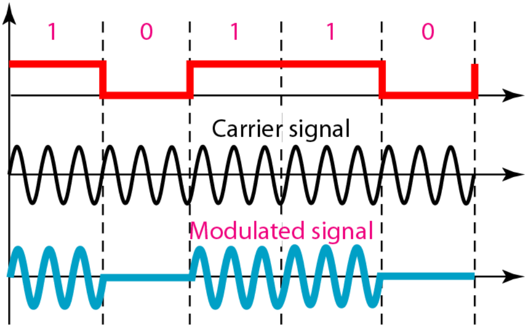
the tradeoff is unavoidable:
Perfectly limited bandwidth causes time spreading.
Perfectly limited time causes wide bandwidth.

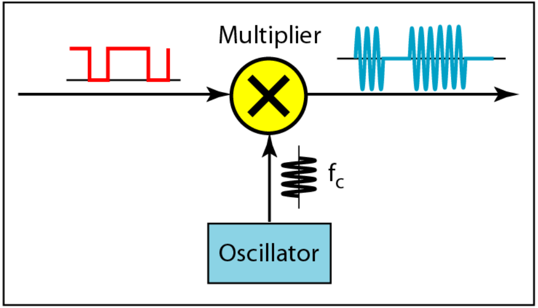
d	Bandwidth	Frequency-domain effect	Time-domain effect
0	S	Abrupt cutoff; ideal minimum bandwidth.	Long pulse tails; exact implementation is unrealistic.
0.5	$1.5S$	Moderate transition band.	More practical pulse shape with less ringing.
1	$2S$	Widest transition band.	Smoothest filtering and shortest ringing, but uses more spectrum.

Engineering decision: Small d saves hertz but needs sharper filtering; large d spends more hertz for easier generation and recovery.

Implementation of Binary ASK

- A multiplier gates/scales the carrier using input bits.
- Output amplitude follows bit pattern.
- Receiver decides based on detected envelope/energy.





Example 5.3 (ASK)

Problem: Available band is 200–300 kHz ($B = 100$ kHz). Find carrier frequency and bit rate for ASK with $d = 1$.

- Carrier at band midpoint: $f_c = 250$ kHz
- $B = (1 + d)S = 2S \Rightarrow S = 50$ kbaud
- For binary ASK, $r = 1 \Rightarrow N = S = 50$ kbps

Result

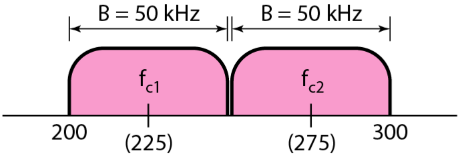
$f_c = 250$ kHz, data rate $N = 50$ kbps.

Example 5.4 (Full-Duplex ASK)

Significance: Full-duplex enables simultaneous bidirectional communication (both sender and receiver transmit at the same time on different frequency bands).

Idea: Full duplex splits available band into two directional channels.

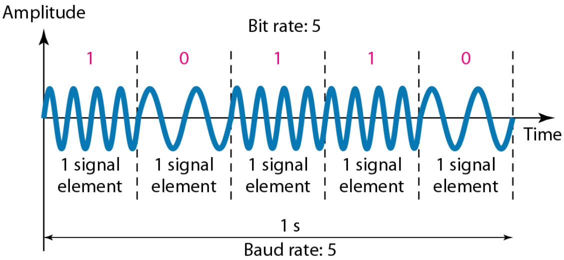
- Total band: 100 kHz
- Per direction: 50 kHz
- With ASK and $d = 1$: $S = 25$ kbaud
- Binary case ($r = 1$): $N = 25$ kbps per direction

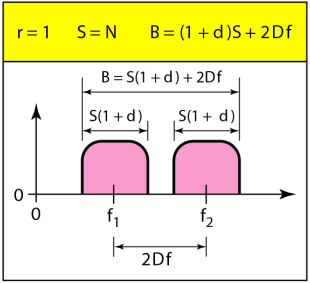


FSK (Frequency Shift Keying)

- Carrier frequency changes with data bit.
- Peak amplitude is constant; information is carried by tone choice, not amplitude.
- Binary FSK uses two tones: f_1 and f_2.
- Tone spacing around midpoint is $\pm\Delta f$, so separation is $2\Delta f$.

Figure: Binary 1 and 0 are mapped to two distinct carrier frequencies.



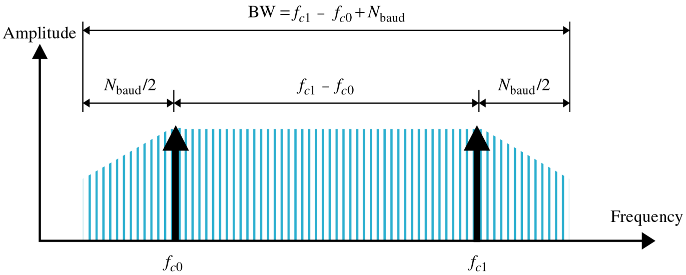


FSK Bandwidth

- FSK can be viewed as two shifted ASK-like spectra.
- Required bandwidth includes symbol shaping and tone separation.
- Textbook model:

$$B_{\mathsf{FSK}} = (1 + d)S + 2\Delta f$$

- Compared with ASK/PSK, BFSK usually needs more bandwidth.



Example 5.5 (FSK)

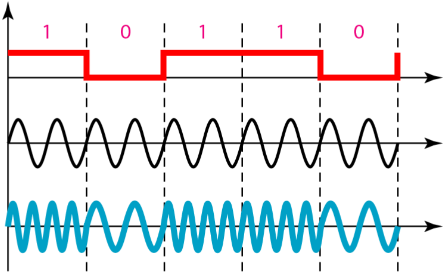
Problem: Available band is 200–300 kHz ($B = 100$ kHz). Use FSK with $d = 1$. Find carrier and bit rate.

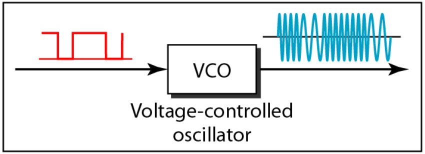
- Midpoint carrier $f_c = 250$ kHz
- Choose $2\Delta f = 50$ kHz
- $B = (1 + d)S + 2\Delta f = 2S + 50$ kHz
- $100 = 2S + 50 \Rightarrow S = 25$ kbaud
- Binary FSK ($r = 1$): $N = 25$ kbps

Figure: Implementation of binary FSK

- Input bits select one of two oscillators.
- Output switches between f_1 and f_2.
- Demodulator detects dominant tone per symbol.

The implementation maps each bit interval to one of two frequencies.





Multilevel FSK (MFSK)

- MFSK uses L (or M) frequencies to send $\log_2 L$ bits per symbol.
- Example: 4 frequencies can send 2 bits per symbol.
- Textbook bandwidth model:

$$B_{\text{MFSK}} = (1 + d)S + (L - 1)2\Delta f$$

- For proper operation, minimum spacing is typically $2\Delta f \geq S$.

Tradeoff

MFSK can improve bit-per-symbol efficiency, but larger L quickly expands bandwidth.

Example 5.6 (MFSK)

In MFSK, each tone is not detected just by “eyeballing the spectrum gap.”
It is detected over one symbol interval, and the tones are chosen to be sufficiently separable/orthogonal over that interval.
Why it looks like the spacing is almost zero
In the figure, each pink “lobe” has width, and neighboring lobes almost touch.
That is because each tone is transmitted for only a finite symbol duration. A finite-duration tone does not appear as a single infinitely thin line in frequency. Instead, it spreads into a small band.
So each tone has: a center frequency f_1,f_2,…,f_8, plus some spectral spread around it. That is why the lobes look close.

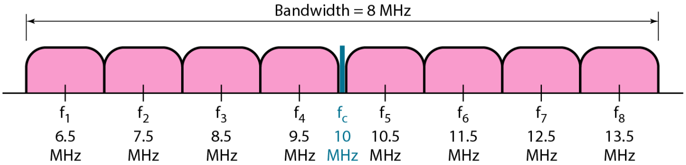
But does that mean the receiver will confuse them?
Not automatically. The receiver is not just checking “is there empty space between the lobes?” It usually checks: “Which of the allowed tones does the received signal match best during this symbol interval?” So if the tone spacing is chosen properly, the receiver can still distinguish them.

FSK is generally safer than ASK but it costs more bandwidth.

Given: $N = 3$ Mbps, 3 bits/symbol, $f_c = 10$ MHz.

- $r = 3 \Rightarrow L = 2^3 = 8$ tones
- $S = \frac{N}{r} = \frac{3\text{ M}}{3} = 1$ Mbaud
- Use $2\Delta f = S = 1$ MHz
- $B = (1 + d)S + (L - 1)2\Delta f = (8 + d)$ MHz
- If $d = 0$, $B = 8$ MHz; if $d = 1$, $B = 9$ MHz

Figure: The figure allocates eight frequencies around the carrier region.

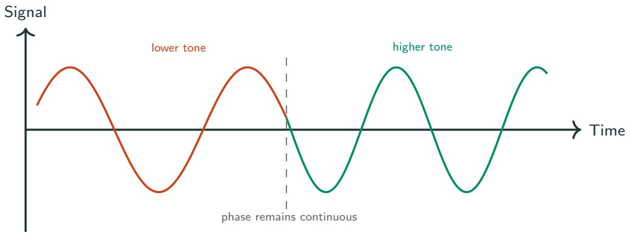


Binary FSK vs M-ary FSK

Scheme	How symbols are represented	Main engineering trade-off
Binary FSK	2 tones, 1 bit/symbol	Simpler implementation, moderate spectral efficiency
M-FSK	M distinct tones ($\log_2 M$ bits/symbol)	Better distance between symbols possible, but bandwidth grows quickly as M increases
Orthogonal FSK design	Tone spacing chosen for near-orthogonality over symbol interval	Reduces detector confusion at the cost of spacing/bandwidth

Takeaway: Large M-FSK can be power-efficient in some regimes, but it is usually bandwidth-hungry (opposite trend of high-order QAM).

MSK Signal: Continuous Phase Transition



GMSK: Gaussian-Filtered MSK (GSM Context)

What changes from MSK

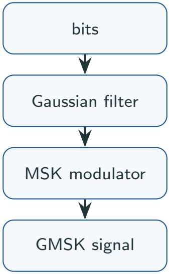
Before frequency modulation, data is passed through a Gaussian filter.

Effect: smoother phase trajectory and tighter occupied spectrum.

System relevance

2G GSM uses GMSK to balance spectral compactness, constant-envelope transmission, and hardware practicality.

GMSK Transmitter Block Diagram



- Gaussian filtering smooths data before modulation
- Narrows occupied spectrum by reducing sidelobes and out-of-band radiation
- Maintains constant envelope for RF efficiency

MSK: Continuous-Phase FSK with Minimum Shift

Main idea and benefit

MSK is a continuous-phase FSK variant with minimum tone spacing that keeps symbols orthogonal over one symbol interval.

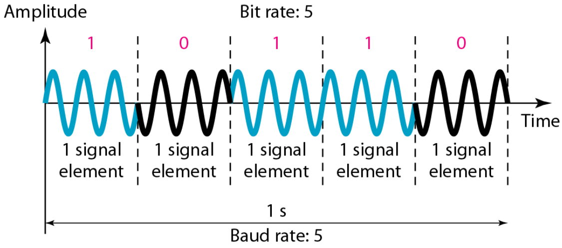
Because phase stays continuous, spectral sidelobes are better controlled than in abrupt discontinuous switching.

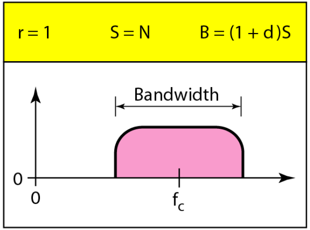
- **Key property:** Phase transitions smoothly from symbol to symbol
- **Benefit:** Narrower spectrum compared to standard FSK
- **Tradeoff:** Slightly more complex receiver

PSK (Phase Shift Keying)

- Carrier phase states represent symbols.
- Amplitude and frequency stay constant.
- Binary PSK uses two phases separated by 180°.
- More phases (M-PSK) can carry more bits per symbol.

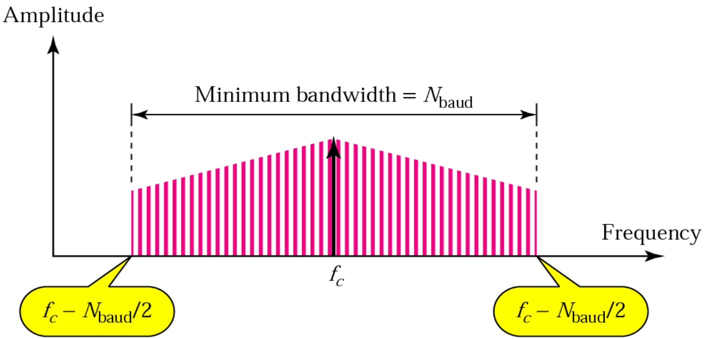
Figure: PSK stores information in carrier phase states.





PSK Bandwidth Relation

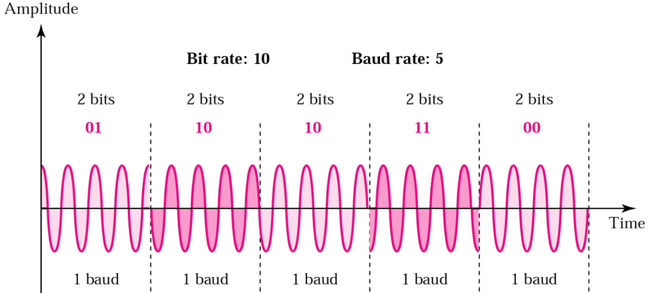
- In the binary case, PSK bandwidth is comparable to binary ASK.
- PSK bandwidth is typically lower than binary FSK bandwidth.
- This is one reason PSK is widely used in spectrally constrained links.



BPSK Mapping and Constellation

- BPSK maps bit 0 and bit 1 to two opposite phases.
- Common mapping: $0 \rightarrow 0^\circ$, $1 \rightarrow 180^\circ$.
- Constellation has two points on the in-phase axis.

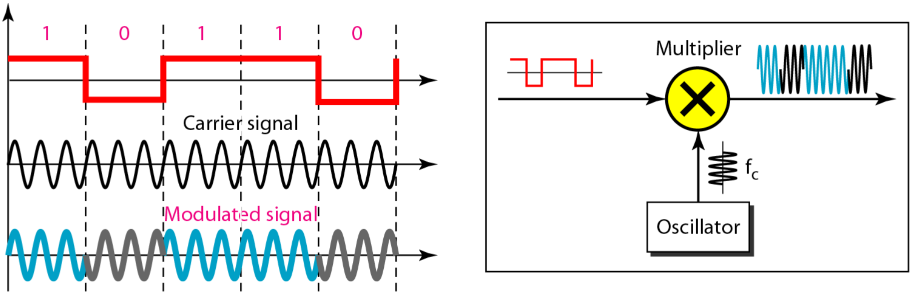
$$s(t) = A_c \cos(2\pi f_c t + \phi_i), \quad \phi_i \in \{0, \pi\}$$



Implementation of BPSK

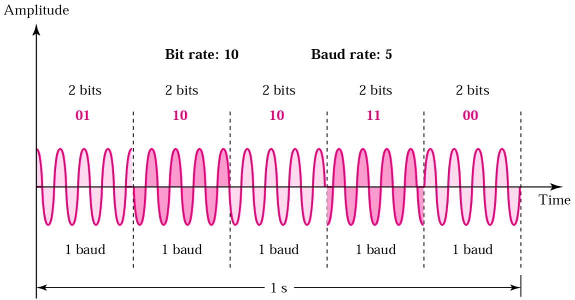
Figure: Implementation of BPSK

- A phase switch toggles the carrier by 180°.
- This creates symbol waveforms for bits 0 and 1.
- Coherent detection recovers phase state at the receiver.



QPSK (4-PSK) Method

- QPSK uses four phase states.
- Each symbol carries 2 bits ($r = 2$).
- For the same bit rate, QPSK halves symbol rate compared with BPSK.
- Lower symbol rate generally reduces required bandwidth.

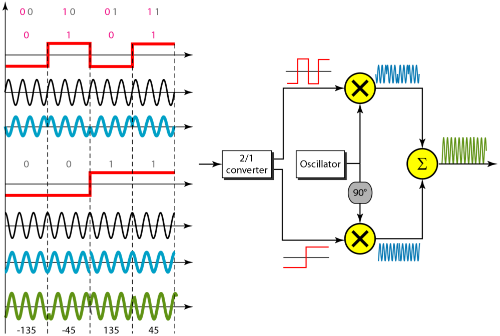


QPSK and Its Implementation

Figure: QPSK and implementation

- Two-bit groups select one of four phases.
- Practical implementations use in-phase and quadrature branches.
- Receiver decides among four constellation quadrants.

QPSK works by taking the input bits in groups of two and using each pair to choose one of four possible phases of a carrier wave. Instead of changing the amplitude or frequency of the carrier, QPSK changes where the sine wave starts in its cycle, which is called its phase. For example, the bit pairs 00, 01, 10, and 11 can each be assigned to a different phase. Since every transmitted symbol represents two bits, QPSK can send the same amount of data using fewer symbol changes than a method that sends only one bit per symbol. At the receiver, the incoming wave’s phase is measured, and the receiver decides which of the four possible phases was sent, then converts that phase back into the corresponding two-bit group.



Where the Minus Sign Comes From

$$s(t) = A\cos(\omega_c t + \theta)$$

$$\cos(a + b) = \cos a \cos b - \sin a \sin b$$

$$s(t) = A\cos\theta\cos(\omega_c t) - A\sin\theta\sin(\omega_c t)$$

- Define $I = A\cos\theta$, $Q = A\sin\theta$.
- Then
$$s(t) = I\cos(\omega_c t) - Q\sin(\omega_c t)$$
- So the negative sign in front of the Q-branch comes directly from the trigonometric expansion.

QPSK Phase from I, Q: Vector View

$$s(t) = I\cos(\omega_c t) - Q\sin(\omega_c t), \qquad I, Q \in \{+1, -1\}$$

- Think of (I, Q) as a point in the IQ plane: I=horizontal axis, Q=vertical axis.
- The carrier phase is the angle of that point measured from $+I$ axis.
- Use $\theta = \text{atan2}(Q, I)$ (equivalently: reference angle $\tan^{-1}(|Q/I|)$ + quadrant correction).
- Here $|Q/I| = 1$, so the reference angle is always 45°.

QPSK chooses two values, I and Q, from +1 or -1. These two values define a point on a 2D plane. The angle of that point determines the phase of the transmitted carrier. Because there are four possible points, QPSK has four possible phases, so it can send two bits per symbol.

QPSK Angles by Quadrant (Worked Cases)

- $I = +1, Q = +1 \Rightarrow (+1, +1)$: first quadrant $\Rightarrow \theta = 45^\circ$
- $I = -1, Q = +1 \Rightarrow (-1, +1)$: second quadrant $\Rightarrow \theta = 180^\circ - 45^\circ = 135^\circ$
- $I = -1, Q = -1 \Rightarrow (-1, -1)$: third quadrant $\Rightarrow \theta = 180^\circ + 45^\circ = 225^\circ \equiv -135^\circ$
- $I = +1, Q = -1 \Rightarrow (+1, -1)$: fourth quadrant $\Rightarrow \theta = -45^\circ$ (or 315°)

Conclusion

So the four QPSK phase values are exactly $45^\circ, 135^\circ, -135^\circ, -45^\circ$.

Example 5.7 (QPSK Bandwidth)

Problem: Find bandwidth for 12 Mbps QPSK with $d = 0$.

- QPSK carries 2 bits/symbol $\Rightarrow r = 2$
- Symbol rate: $S = \frac{N}{r} = \frac{12 \text{ Mbps}}{2} = 6$ Mbaud
- For $d = 0$, $B = S = 6$ MHz

Result

Required bandwidth is 6 MHz.

Differential PSK (DPSK): Avoid Absolute-Phase Tracking

Main idea

Encode information in **phase change** between consecutive symbols, not in absolute phase.

Example (binary DPSK): bit $0 =$ no phase change, bit $1 = \pi$ phase flip

Why use it

Receiver can avoid strict carrier-phase reference recovery. Useful when coherent phase tracking is difficult or costly. Tradeoff: usually a modest BER penalty versus coherent PSK.

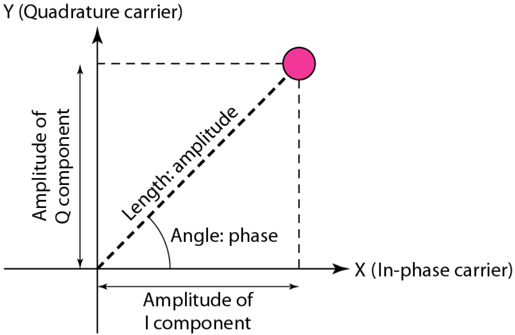
In ordinary PSK, the receiver must know the exact reference phase of the carrier. For example, it must know what “0∘” and “180∘” mean. If the receiver’s carrier reference is shifted, it may decode the bits wrongly. DPSK avoids needing such strict absolute phase tracking because it compares each received symbol with the previous received symbol. It only asks: Did the phase stay the same, or did it change? So DPSK is useful when accurate carrier phase recovery is difficult, expensive, or unreliable. The tradeoff is the BER penalty. BER means bit error rate. DPSK is usually a little more error-prone than coherent PSK because it depends on comparing two consecutive symbols. If noise affects one symbol, it can affect the phase-change decision between symbols.

Where this appears

Differential phase ideas are widely used in practical systems for robust synchronization startup and low-complexity detection modes.

Constellation Diagram Concept

Figure: Concept of a constellation diagram



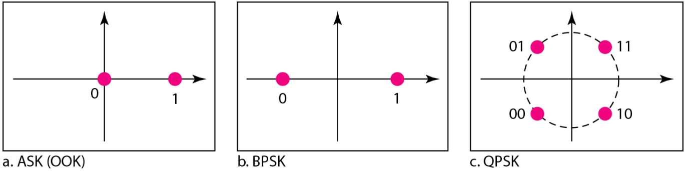
- A constellation point represents one symbol.
- Horizontal axis: in-phase component (I)
- Vertical axis: quadrature component (Q)
- Point spacing affects noise tolerance and error rate.

Example 5.8: ASK, BPSK, QPSK Constellations

Task: Draw constellation diagrams for OOK-ASK, BPSK, and QPSK.

- OOK: two amplitude states on one axis
- BPSK: two opposite phase points
- QPSK: four quadrant points

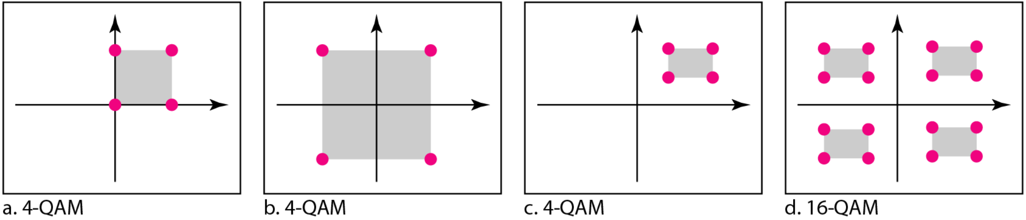
Figure: Compares symbol geometry across three schemes



QAM (Quadrature Amplitude Modulation)

- QAM varies both amplitude and phase of the passband signal.
- Equivalent implementation: choose amplitudes on in-phase (I) and quadrature (Q) carriers.
- Enables high data rates with multi-level constellations.
- Larger constellations increase noise sensitivity.

Figure: QAM constellation example

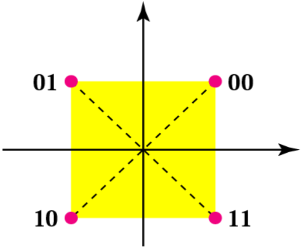


Digital-to-Analog Conversion: QAM View

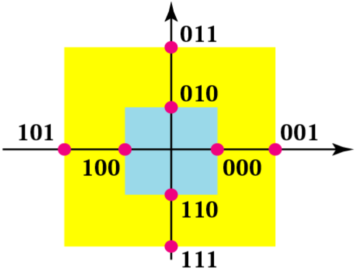
- QAM transmits symbols by selecting (I, Q) amplitudes.
- Equivalent passband form:

$$s(t) = I(t)\cos(2\pi f_c t) - Q(t)\sin(2\pi f_c t)$$

- Modern wireless and cable systems rely heavily on QAM.

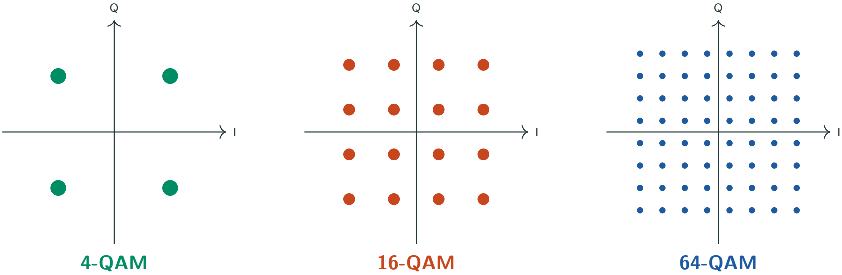


4-QAM
1 amplitude, 4 phases



8-QAM
2 amplitudes, 4 phases

Constellation Family: 4-QAM, 16-QAM, 64-QAM



Visual trend: higher throughput means denser point packing, so cleaner channels are required.

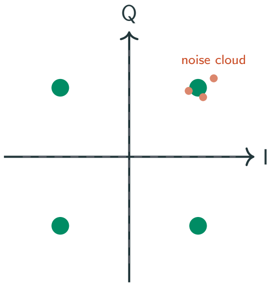
Spectral-Efficiency Comparison at a Glance

Scheme	Bits/symbol	Envelope	Typical relative behavior	Common usage
BPSK	1	Constant	Very robust, low spectral efficiency	Control channels, deep-fade links
QPSK	2	Constant	Good robustness-efficiency balance	Satellite, LTE/Wi-Fi base modes
16-QAM	4	Varying	Higher throughput, needs better SNR	LTE/Wi-Fi normal data modes
64/256-QAM	6/8	Varying	Very high throughput, SNR-hungry	Good-channel broad-band operation

Not a single winner

The best modulation depends on channel quality, target data rate, and power constraints.

Constellation Diagram: Receiver Decision Geometry



Interpretation

Ideal symbols are points. Real received samples are scattered clouds around them because of noise, interference, and channel impairments. Detector chooses the nearest ideal point (minimum-distance decision).

Distance matters: larger point spacing means less cloud overlap and lower error probability.

Distance, Noise, and Error Probability

For AWGN channels, uncoded BER intuition can be summarized as:

$$\text{better BER} \iff \text{larger minimum symbol distance } d_{\min}$$

$$\text{and/or larger } E_b/N_0$$

E_b/N_0 meaning

E_b: energy used to send one bit
N_0: noise spectral density
Higher E_b/N_0 usually pushes BER downward

Engineering meaning

If channel quality drops, systems often lower modulation order (e.g., 64-QAM $\rightarrow$ QPSK) so points are farther apart and BER remains acceptable.

BER and SER: Reliability Metrics

$$\text{BER} = \frac{\#\ \text{bit errors}}{\#\ \text{transmitted bits}}, \qquad \text{SER} = \frac{\#\ \text{symbol errors}}{\#\ \text{transmitted symbols}}$$

BER is usually the system-level metric for user data reliability.

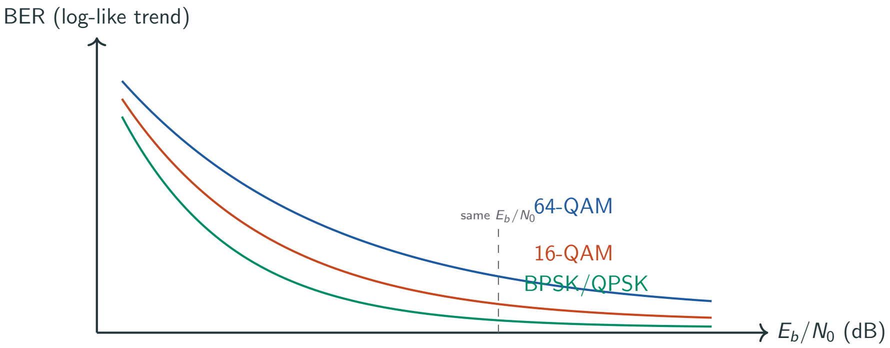
What increases BER

Lower E_b/N_0, Denser constellations at fixed power, Poor synchronization and channel estimation, Interference/fading

What decreases BER

Higher received bit energy, Better estimation/equalization, Stronger coding (Later Topic), Conservative modulation under poor channels

Qualitative BER Curves vs E_b/N_0



Interpretation

At the same E_b/N_0, higher-order constellations generally have higher BER. To use them safely, systems need cleaner channels or stronger coding.

Unified Tradeoff View

If you increase...	Then usually...
Modulation order M	Throughput increases, but required SNR for target BER increases
Symbol spacing (for fixed noise)	BER improves, but either power or bandwidth cost appears elsewhere
FSK tone spacing	Detection robustness improves, but occupied bandwidth increases
Coding strength (Later Topic)	BER improves, but useful payload rate may drop for fixed raw symbol rate

Engineering mindset

There is no universally best modulation. Design means selecting an operating point under power, bandwidth, hardware, and reliability constraints.

Where Each Modulation Family Appears

Family	Typical systems	Why used there	Typical concern
ASK/OOK	Optical links, simple RF keying	Very simple transmitter/receiver	Sensitive to amplitude variation
FSK/M-FSK	Telemetry, low-rate radios, IoT links	Robust noncoherent options available	Bandwidth growth with tone count
PSK/DPSK	Satellite, wireless control/data channels	Constant-envelope friendliness, good power efficiency	Phase synchronization complexity
QAM	Wi-Fi, LTE/5G, cable and DSL systems	High spectral efficiency for high throughput	Needs high SNR and linear RF chain
MSK/GMSK	GSM and constant-envelope radio designs	Continuous phase, compact spectrum	Lower peak throughput than dense QAM modes

Common Confusions and Quick Fixes

Common confusion	**Correct interpretation**
"Bit rate and symbol rate are the same."	Only for binary signaling. In general, $R_s = R_b / \log_2 M$.
"Higher-order modulation is always better."	Better throughput, yes; but BER robustness worsens at fixed power/noise.
"Constellation is just a drawing."	It is the receiver's decision geometry and directly predicts error behavior.
"DPSK is just PSK with a new name."	DPSK changes what is detected (phase *difference*, not absolute phase), reducing synchronization burden.
"MSK/GMSK are unrelated to FSK."	They are continuous-phase members of the FSK family.

Summary

- Digital modulation maps bits to carrier waveforms so passband channels can carry digital information
- ASK, FSK, and PSK families differ by which carrier parameter is changed
- M-ary signaling raises bits/symbol but tightens decision spacing for fixed power
- Constellation diagrams provide direct geometric intuition for symbol decisions and BER trends
- QAM dominates high-throughput links because of spectral efficiency; adaptive modulation manages its SNR demands
- MSK/GMSK improve spectral behavior in constant-envelope systems
- BER/SER and E_b/N_0 connect physical channel quality to modulation choice

References

1. B. A. Forouzan, *Data Communications and Networking*, 5th ed., Ch. 5 digital-to-analog conversion and digital modulation (ASK, FSK, PSK, QAM, M-ary concepts), commonly around pp. 151–166 depending printing.
2. A. Goldsmith, *Wireless Communications*, Ch. 5 (pp. 126–171): passband digital communication, constellation thinking, and BER/SNR tradeoffs.
3. T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed., Ch. 5–6: modulation families, receiver viewpoint, and practical wireless context (page numbering varies across printings).

Exact pagination can vary across international printings.

Appendix — First-Exposure Terminology I

Term	Meaning in this topic
Passband digital signal	Digitally modulated waveform centered around a nonzero carrier frequency
Symbol interval (T_s)	Time duration used to transmit one modulation symbol
Symbol rate (R_s)	Number of symbols per second (baud)
Modulation order (M)	Number of distinct symbol choices in a signaling scheme
Constellation	Set of allowed symbol points in the I-Q plane
Decision boundary	Region separator used by receiver to map noisy points to nearest symbol
Gray coding	Bit labeling where adjacent symbols differ by one bit
Coherent detection	Detection using a phase/frequency-aligned carrier reference
Noncoherent detection	Detection that avoids strict carrier phase reference (common in some FSK/DPSK modes)

Appendix — First-Exposure Terminology II

Term	Meaning in this topic
Minimum distance ($d_{\min}$)	Smallest spacing between any two constellation points
BER / SER	Bit Error Rate / Symbol Error Rate reliability measures
E_b/N_0	Bit-energy to noise-density ratio; key SNR-like metric for digital links
Spectral efficiency (η)	Useful bit rate per Hz, typically bits/s/Hz
Constant-envelope modulation	Modulation with nearly fixed amplitude (e.g., PSK/FSK families)
Differential encoding	Information mapped to phase changes between symbols
MSK	Minimum Shift Keying: continuous-phase orthogonal FSK variant
GMSK	Gaussian-filtered MSK used in GSM-style systems
Adaptive modulation	Dynamic modulation-order switching based on channel quality feedback

Appendix — Source Mapping I

Slide-to-source mapping (page ranges follow commonly used editions; local printings may shift slightly).

Slide portion	Primary source	Source pages used
Learning objectives **Why carrier modulation** **Base-band vs. band-pass** **Channel mismatch** **Carrier/demodulation math**	Forouzan + Goldsmith + instructor synthesis	Forouzan Ch. 3 signal/bandwidth ideas and Ch. 5 digital-to-analog conversion opening (commonly Ch. 5 pp. 151–154); Goldsmith Ch. 5, pp. 126–136 for passband/baseband framing. Fourier and demodulation derivations are instructor-synthesized.
Digital-to-analog conversion **Data rate vs. signal rate** **Examples 5.1–5.2**	Forouzan	Forouzan Ch. 5 digital-to-analog conversion preliminaries, bit rate/baud rate, and signal-element examples (commonly pp. 151–154).
ASK waveform **ASK bandwidth** **Baud-to-bandwidth explanation** **Pulse shaping / roll-off** **Raised cosine / filtering tradeoff** **Examples 5.3–5.4**	Forouzan + Goldsmith + instructor synthesis	Forouzan Ch. 5 ASK and bandwidth model $B = (1 + d)S$ (commonly pp. 154–157); Goldsmith Ch. 5, pp. 126–136 for pulse-shaping/passband intuition. Added unit-comparison and filtering-tradeoff slides are explanatory synthesis.

Appendix — Source Mapping II

Slide portion	Primary source	Source pages used
FSK intuition + BFSK waveform **FSK bandwidth** **Examples 5.5–5.6** **Binary vs. *M*-FSK**	Forouzan	Forouzan Ch. 5 FSK and multilevel FSK sections (commonly pp. 157–159 depending printing).
PSK intuition + bandwidth **BPSK mapping / implementation** **QPSK method + implementation** ***M*-PSK tradeoff** **DPSK**	Forouzan + Goldsmith + Rappaport	Forouzan Ch. 5 PSK family (commonly pp. 159–162); Goldsmith Ch. 5, pp. 136–152; Rappaport Ch. 5–6 conceptual receiver discussion (section-page numbering varies by printing).
Constellation concept **ASK/BPSK/QPSK examples** **Decision geometry** **Distance, noise, error probability**	Goldsmith + Rappaport + instructor synthesis	Goldsmith Ch. 5, pp. 136–171 for signal-space and E_b/N_0 behavior; Rappaport Ch. 5–6 receiver and error-performance context. Figures and decision-region explanations are redrawn/synthesized for first exposure.

Appendix — Source Mapping III

Slide portion	Primary source	Source pages used
QAM equation and constellation family		
Spectral-efficiency comparison		
Adaptive modulation	Forouzan + Goldsmith + Rappaport	Forouzan Ch. 5 QAM and M-ary sections (commonly pp. 162–166); Goldsmith Ch. 5, pp. 152–171 for QAM/spectral-efficiency tradeoffs; Rappaport Ch. 5–6 for wireless adaptation context.
MSK slide		
Continuous phase transition		
GMSK slides	Forouzan + Rappaport + synthesized bridge	Forouzan Ch. 5 MSK/GMSK-family placement near the end of the digital-modulation sequence; Rappaport Ch. 6 practical wireless modulation context. GSM-specific framing is synthesized at conceptual level to fit course prerequisite limits.
BER/SER metrics		
Qualitative BER curves		
Unified tradeoff view	Goldsmith + Rappaport + instructor synthesis	Goldsmith Ch. 5 BER/SNR trends (pp. 145–171); Rappaport Ch. 5–6 receiver-performance context (edition-dependent pages). Qualitative curves and tradeoff table are redrawn for teaching intuition.
System usage table		
Common confusions		
Summary + terminology appendix	Forouzan + Goldsmith + Rappaport + instructor synthesis	Integrated from cited chapters to connect textbook concepts with this course's Topic 6/7/10 flow and student-first explanations.